

CAC

Cellular Automata Checkerboards

Cian, Azul

September 2026

1 Introduction

The idea behind CACs is to explore the relationship between different CA rules. We achieve this by generalising CAs with checkerboard patterns with N rules in the same board.

2 Definition

We will denote CACs with the uppercase letter C. If needed, subscripts may be used: $C_1, C_2, C_3, \dots, C_N$ are all specific CACs.

We will denote board sizes with a subscript too. For example, $C_{n \times m}$ denotes a CAC named C with n number of columns and m number of rows.

We will denote a cell with the lowercase letter c. Here, the subscripts will represent a position.

For example, c_{ij} is the cell at row i, column j (in a given CAC C, presumably).

Let C_{AB} be a CAC of rules R_A and R_B .

Then:

$$C_{AB} = \{ \forall i, j : (0 \leq i < n) \wedge (0 \leq j < m) \wedge \\ (i + j \equiv 0(2) \iff c_{i+j} \in R_A) \wedge \\ (i + j \equiv 1(2) \iff c_{i+j} \in R_B) \}$$

[1] We'll call the given definition order 2. Note that you can generalise CACs to order N (N rules) by extending the definition to mod N (instead of only using mod 2).

[2] Note that you can generalise CACs to N dimensions by using N indices.

Cardinal

We define $\#C_A$ (spelt as "the cardinal of rule A in C") as the total number of cells subject to rule A in CAC C.

That is to say:

$$\#C_{n \times m_A} = \sum_{i=1}^n \sum_{j=1}^m c_{ij} \quad \forall c \quad / \quad (c_{ij} \in R_A)$$

3 Properties

Let C be a CAC in a board of size $n \times m$ and rules $\mathbb{R} = \{A, B, \dots, Z\}$.

Then:

$$\lim_{\substack{m \rightarrow \infty \\ n \rightarrow \infty}} \#A = \#B \quad \forall A, B \in \mathbb{R}$$

3.1 Inverse

The inverse of C_{AB} is C_{BA} :

$$C_{AB}^{-1} = C_{BA} \quad \forall R_A, R_B$$

3.2 Bounding box

We can look for the bounding box of a pattern (configuration) by looking for its exact boundaries.

That is to say, for a given bounding box B , both of these two conditions are met:

[1] There is no lower bound bigger than B .

and

[2] There is no upper bound smaller than B .

(A more formal definition is still WIP)

4 Future

Desired structures yet to discover:

- * Logic gates (AND, OR, NOT...)
- * Universal constructors